SHUBHAM PANDEY

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Summary

AI/ML Engineer and final-year B.Tech student specializing in Artificial Intelligence and Software Engineering. Strong foundation in Data Structures & Algorithms (DSA) and Object-Oriented Programming (OOP) with hands-on experience building production-ready AI agent architectures, enterprise Retrieval-Augmented Generation (RAG) pipelines, and high-performance Computer Vision applications. Proficient in LLM integration, multi-agent orchestration (LangChain, LangGraph, CrewAI), and scalable system architecture. Seeking software engineering or AI/ML roles to drive intelligent automation and scalable AI product development.

Education

Bennett University — 2023 – 2027
Bachelor of Technology (B.Tech) in Computer Science Engineering — CGPA: 8.0 / 10.0

Shiksha Bharati Senior Secondary School (CBSE) — 2022 – 2023
Class XII — 84% / 100

Shiksha Bharati Senior Secondary School (CBSE) — 2020 – 2021
Class X — 91% / 100

Skills

AI/ML & Agentic Frameworks: LangChain, LangGraph, CrewAI, RAG Pipelines, Prompt Engineering, Multi-Agent Systems, LLM Integration, Groq API, ChromaDB, FAISS, Vector Databases

Deep Learning & Machine Learning: PyTorch, TensorFlow, Scikit-Learn, Neural Networks, NLP, Transfer Learning, Object Detection (YOLO), Computer Vision, GANs, Model Fine-Tuning

Data & Analytics: Pandas, NumPy, TF-IDF, Cosine Similarity, Spline Interpolation, Data Preprocessing, Feature Engineering

Programming Languages: Python, C++, JavaScript, TypeScript

Web Frameworks & Deployment: FastAPI, Flask, React, Vite, Tailwind CSS, Streamlit, Uvicorn, REST APIs, Asynchronous Programming (AsyncIO, HTTPX)

Databases, DevOps & Core CS: MongoDB, SQL, Git, GitHub, Docker, CI/CD, Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), System Design

Experience

BCG X (Boston Consulting Group) — GenAI Job Simulation — View Certificate | 06/2026
Generative AI Engineer

- Architected a production-ready corporate financial chatbot by ingesting and parsing unstructured regulatory disclosures using Pandas, reducing document review time by 30% for enterprise stakeholders
- Extracted and structured critical business performance metrics, operational risk signals, and KPI analytics from raw filings, improving data accessibility by 25% for downstream AI reasoning tasks
- Engineered context-aware prompt workflows and modular system prompts tailored for scalable enterprise language models, boosting system alignment metrics by 15%

JPMorgan Chase & Co. — Quantitative Research Job Simulation — View Certificate | 06/2026
Quantitative Research Analyst

- Developed a custom cubic spline interpolation model in Python to forecast commodity prices across a 12-month forward horizon, demonstrating applied quantitative finance and time-series modeling skills
- Designed a reusable asset valuation framework to dynamically estimate baseline asset configurations, reducing manual evaluation overhead by 20% and enhancing contractual profitability metrics
- Optimized financial risk evaluation pipelines by implementing programmatic borrower segmentation using credit score bucketing, accelerating risk assessment workflows by 15%

Projects

Enterprise Multi-Agent Platform
GitHub | 04/2026

Core Frameworks: Python, React, TypeScript, Web Services, Multi-Agent Systems, System Design

- Architected a modular multi-agent orchestration system integrating 4 enterprise domains (Business, Finance, Sales, HR) via advanced state machines, accelerating cross-departmental data synthesis by 30%
- Implemented dedicated asynchronous microservices to isolate business logic per domain and successfully handle 500+ concurrent operational requests
- Delivered an executive-facing React dashboard with real-time analytics, structured financial forecasting tools, and budget modules, improving reporting efficiency by 25%

Document Intelligence Engine
GitHub | 02/2026

Core Frameworks: Python, React, Vector Databases, Cloud APIs, Large Language Models, Semantic Search

- Built a production-grade Retrieval-Augmented Generation system supporting 6 distinct file types including digital documents and spreadsheet formats, serving as a scalable enterprise question-answering engine
- Optimized database search and retrieval performance with language models, achieving an end-to-end latency of 0.69 seconds and a retrieval latency of 16.6 milliseconds — a 40%+ improvement over baseline configurations
- Enforced local embedding generation with structural chunk mapping to guarantee 100% citation-backed, hallucination-resistant responses

Conversational Discovery Application
GitHub | 03/2026

Core Frameworks: Python, React, Machine Learning, Natural Language Processing, Parallel Pipelines

- Developed a semantic discovery engine using advanced vectorization, cosine similarity, and tokenization features to map natural language query intents to relevant database metadata
- Programmed real-time metadata enrichment by implementing asynchronous parallel pipelines, reducing query latency by 35% in live workflows
- Integrated a conversational layer powered by cloud language models alongside fuzzy matching algorithms for typo-tolerance, enhancing user engagement metrics by 20%

Traffic Sign Recognition System
GitHub | 01/2026

Core Frameworks: Python, React, Computer Vision, Deep Learning, Deep Neural Networks (DNN)

- Trained and fine-tuned a custom object detection model achieving a verified 99.05% accuracy on sign recognition patterns, demonstrating expert-level computer vision model development
- Constructed hardware-accelerated inference pipelines processing image, video, and live webcam streams with a latency of 19 to 25 milliseconds, enabling real-time edge-compatible deployment
- Integrated browser-native high-performance video output via advanced compression encoding with interactive interfaces, improving processing efficiency by 30%

Personal Portfolio
Live Project | GitHub | 06/2026

Core Frameworks: React, Vite, TypeScript, Tailwind CSS, Interface Animation, Version Control

- Designed and deployed a production-ready interactive portfolio on Vercel, serving as a live technical showcase with project deployment tracking, dynamic logs, and responsive UI across all devices

Achievements & Mentorship

- **Technical Peer Mentorship:** Mentored and guided junior engineering cohorts in core Computer Science fundamentals (DSA, OOP) and AI/ML technologies, effectively accelerating their technical problem-solving and software development capabilities

Certifications

Apply Generative Adversarial Networks (GANs) — DeepLearning.AI	View Certificate
Build Basic Generative Adversarial Networks (GANs) — DeepLearning.AI	View Certificate
Neural Networks and Deep Learning — DeepLearning.AI	View Certificate
Machine Learning: Classification — University of Washington	View Certificate
Statistics for Data Science with Python — IBM	View Certificate